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Exercise 3 - Analysis of Coupled Systems

Ramon Aerne

March 6, 2017

Let's start with writing the problem in terms of a G-Network. We have

$$P = 0 \tag{1}$$

since a customer will either leave the cue or become a negative customer. We have

$$Q_{12} = Q_{21} = 1, \tag{2}$$

since every negative customer triggers a change of queue if it is not empty. And we have

$$P_{12}^- = p_1, P_{21}^- = p_2 \tag{3}$$

with P_{ij}^- the probability that a customer of queue i will go as a negative customer to queue j. Finally we have $D_1 = D_2 = 0$ meaning a trigger cannot remove customers completely out of the network and $d_1 = (1 - p_1), d_2 = (1 - p_2)$ the probabilities that with d_i a customer in queue i will leave the network. Since we want to find the steady state equation we set

$$\frac{d}{dt}p(m, n, t) = 0 \tag{4}$$

where $p(m, n, t)$ is the joint probability distribution at time t for $m, n \geq 0$. With $\lim_{t\to\infty} p(m, n, t) = p(m, n)$ we can write

$$\begin{aligned}\frac{d}{dt}p(m,n) = &- p(m,n)[\lambda_1 + \lambda_2 + \mu_1 I(m > 0) + \mu_2 I(n > 0)] \\ &+ p(m+1,n)\mu_1(1-p_1) + p(m,n+1)\mu_2(1-p_2) \\ &+ p(m-1,n)\lambda_1 I(m > 0) + p(m,n-1)\lambda_2 I(n > 0) \\ &+ p(m+1-1,n+1)\mu_1 p_1 I(m > 0) + p(m+1,n)\mu_1 p_1 I(n = 0) \\ &+ p(m+1,n+1-1)\mu_2 p_2 I(n > 0) + p(m,n+1)\mu_2 p_2 I(m = 0)\end{aligned}$$

with $I(X) = 1$ if X is true, else it is zero. The negative term is the probability of leaving state $p(m, n)$, the others are the probability of entering it in all possible events that can happen (as seen in the lecture).

You should define them the terms in your report

No attempt to describe these events

Traffic equation For our problem the traffic equations are

$$\Lambda_i^+ = \lambda_1 + \sum_{j,l} q_j q_l \mu_j P_{jl}^- Q_{li}$$
$$\Lambda_i^- = \sum_j q_j \mu_j P_{ji}^-$$

since $j \neq k, k \neq i$ this simplifies to

$$\Lambda_1^+ = \lambda_1 + q_1 q_2 \mu_1 p_1, \qquad \Lambda_2^+ = \lambda_2 + q_1 q_2 \mu_1 p_2,$$
$$\Lambda_1^- = \mu_2 p_2 q_2, \qquad \Lambda_2^- = \mu_1 p_1 q_1 =$$

and what we try to prove is that

$$p(m,n) = q_1^m (1-q_1) q_2^n (1-q_2), \quad q_i = \frac{\Lambda_i^+}{\mu_i + \Lambda_i -} \tag{5}$$

is a solution to the steady state for this problem. We rewrite the steady state by taking the negative term to the LHS and dividing by $p(m,n)$ on both sides

$$\begin{aligned}[\lambda_1 + \lambda_2 + \mu_1 I(m>0) + \mu_2 I(n>0)] =& q_1\mu_1(1-p_1) + q_2\mu_2(1-p_2) \\ &+ \frac{\lambda_1}{q_1} I(m>0) + \frac{\lambda_2}{q_2} I(n>0) \\ &+ q_2\mu_1 p_1 I(m>0) + q_1\mu_1 p_1 I(n=0) \\ &+ q_1\mu_2 p_2 I(n>0) + q_2\mu_2 p_2 I(m=0)\end{aligned}$$

we rewrite $q_2\mu_1 p_1 = \frac{1}{q_1}(\Lambda_i^+ - \lambda_1)$ the same with the exchanged indices. and in the same step $q_i = \frac{\Lambda_i^+}{\mu_i + \Lambda_i^-}$, again with both indices resulting in

$$\begin{aligned}[\lambda_1 + \lambda_2 + \cancel{\mu_1 I(m>0)} + \cancel{\mu_2 I(n>0)}] =& q_1\mu_1(1-p_1) + q_2\mu_2(1-p_2) \\ &+ \frac{\cancel{\lambda_1}}{\cancel{q_1}} I(m>0) + \frac{\cancel{\lambda_2}}{\cancel{q_2}} I(n>0) \\ &+ (\cancel{\mu_1} + \Lambda_1^- - \frac{\cancel{\lambda_1}}{\cancel{q_1}}) I(m>0) + q_1\mu_1 p_1 I(n=0) \\ &+ (\cancel{\mu_2} + \Lambda_2^- - \frac{\cancel{\lambda_2}}{\cancel{q_2}}) I(n>0) + q_2\mu_2 p_2 I(m=0)\end{aligned}$$

which simplifies to

$$\begin{aligned}\lambda_1 + \lambda_2 = & q_1\mu_1(1-p_1) + q_2\mu_2(1-p_2) \\ &+ \Lambda_1^- I(m>0) + q_1\mu_1 p_1 I(n=0) \\ &+ \Lambda_2^- I(n>0) + q_2\mu_2 p_2 I(m=0)\end{aligned}$$

separating the first term and again using the traffic equation so that $q_1\mu_1 - q_1\mu_1 p_1 I(n>0) - q_1\mu_1 p_1 I(n=0) = q_1\mu_1 - \Lambda_2^- I(n>0) - q_1\mu_1 p_1 I(n=0)$, similarly for the second term yields

$$\begin{aligned}\lambda_1+\lambda_2 = q_1\mu_1 - \Lambda_2^- I(n>0) - q_1\mu_1 p_1 I(n=0)\\ +q_2\mu_2 - \Lambda_1^- I(m>0) - q_2\mu_2 p_2 I(m=0)\\ +\Lambda_1^- I(m>0) + q_1\mu_1 p_1 I(n=0)\\ +\Lambda_2^- I(n>0) + q_2\mu_2 p_2 I(m=0)\end{aligned}$$

which obviously cancels out most term to result in

$$\lambda_1+\lambda_2 = q_1\mu_1 + q_2\mu_2$$

which is satisfied for two independent M/M/1 queues where $q_i \overset{*}{=} \frac{\lambda_i}{\mu_i}$ holds and therefore we have proven that Eq. 5 is satisfied for this problem.

* holds because if the same analysis (traffic equation) is applied to a M/M/1, we see that (because $p=0$)

$$\Lambda^+_{M/M/1} = \lambda, \quad \Lambda^-_{M/M/1} = 0$$

and therefore

$$q = \frac{\Lambda^+}{\mu+\Lambda^-} = \frac{\lambda}{\mu}$$

to show that the traffic equations hold for M/M/1 we can write the steady state as

$$p(m)\left[\lambda+\mu\, I(m>0)\right] = p(m+1)\mu + p(m-1)\lambda\, I(m>0)$$

which after plugin in the traffic equations leads to

$$\lambda+\mu I(m>0) = q\mu + \underbrace{\frac{\lambda}{q}}_{\mu}\lambda\, I(m>0)$$

$$\lambda = q\mu$$

which again is the definition of q for M/M/1

No references